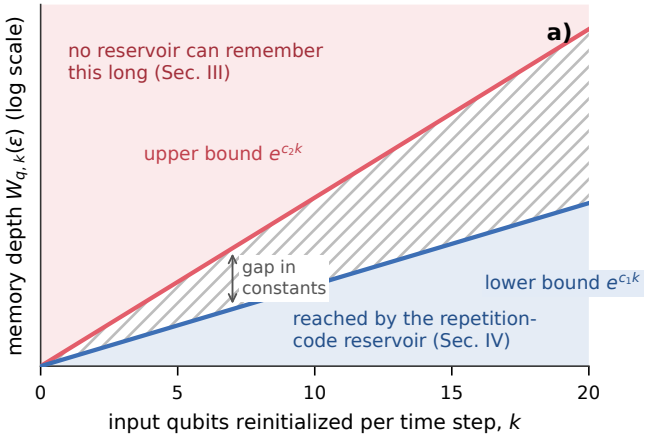
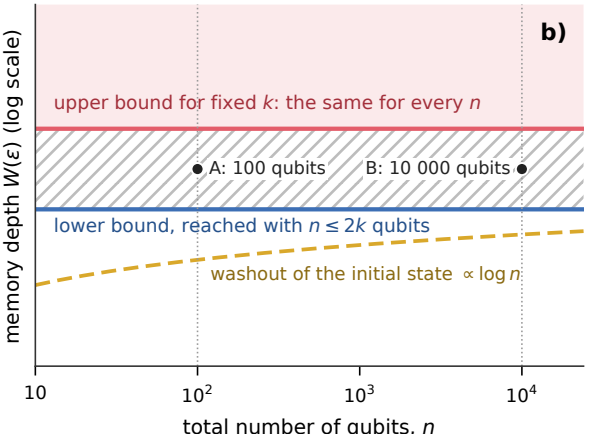

a)
no reservoir can remember this long (Sec. III)
upper bound $e^{c_2 k}$
gap in constants
lower bound $e^{c_1 k}$
reached by the repetition-code reservoir (Sec. IV)
memory depth $W_{q,k}(\varepsilon)$ (log scale)
0
5
10
15
20
input qubits reinitialized per time step, k


b)
upper bound for fixed k: the same for every n
A: 100 qubits
B: 10 000 qubits
lower bound, reached with $n \leq 2k$ qubits
washout of the initial state $\propto \log n$
memory depth $W(\varepsilon)$ (log scale)
10
10^2
10^3
10^4
total number of qubits, n